howest hogeschool

Ergo 3: Virtual Globetrotter

Functional Analysis

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Viola

Scrum Master
AI
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Project Description

Key question

How can we let children with certain limitations experience the world virtually using AI?

Solution

The *Virtual Globetrotter* project is designed to provide children aged 11 to 14 with an opportunity to explore the world virtually through an immersive experience. Using the immersive room at *The Core* at Howest University of Applied Sciences, children can embark on interactive virtual trips that combine AI-generated images and speech recognition to create an engaging and dynamic experience. The images are accompanied by ambient sounds relevant to the scene, enriching the overall experience.

The possibilities for exploration are endless, as **new destinations and experiences can be constantly added**, allowing children to keep discovering and learning about the world in a fun and interactive way.

Target audience analysis

Initially, the project aimed to replicate the experience of a real study trip through AI-generated content. However, after discussions with Ilse Meerschaert, Lector in Ergotherapy at Howest and the client, it was acknowledged that current free AI models cannot accurately replicate real-world locations or architectural details.

The primary educational goal is since then no longer to provide a replacement of the real study trip, but to help children to **think critically about the reliability and accuracy of AI-generated content** and to open the starting point for a conversation with the teacher about the limitations and the advantages in AI and to develop critical thinking and digital literacy by comparing generated images to their expectations or to real-world representations.

While facilitating conversations between children and their teachers is beyond the project's scope, the *Virtual Globetrotter* provides a starting point for meaningful discussions about AI and its role in education.

Minimum Viable Product

MVP features

1. **Continent Selection & Voice Input**:
   - Users can select a continent via a navigation screen.
   - An optional voice input feature enhances the immersive experience.
2. **Immersive Visuals**:
   - Images are displayed seamlessly across all four walls of the immersive room.
   - Images must be high resolution, appropriately scaled, and accurately sized.
3. **Audio Experience**:
   - Relevant sounds or music must accompany the images to heighten immersion.
4. **AI-Generated Images**:
   - All images must be AI-generated.
   - Pregenerated images should match the selected continent and voice input.
   - On-the-spot image generation must occur within an acceptable timeframe.
   - The AI model used for image generation must be free for non-commercial use.

Nice-to-have features:

1. **Alternative Interface**:
   - Allow users to interact with the system via a tablet or phone instead of a keyboard.
2. **Advanced Image Search**:
   - Provide full-text search with fuzziness to locate pregenerated images that approximate the voice input.
3. **AI-Generated Audio**:
   - Leverage AI to create custom music or ambient sounds for enhanced user engagement.

Edge Cases and Solutions

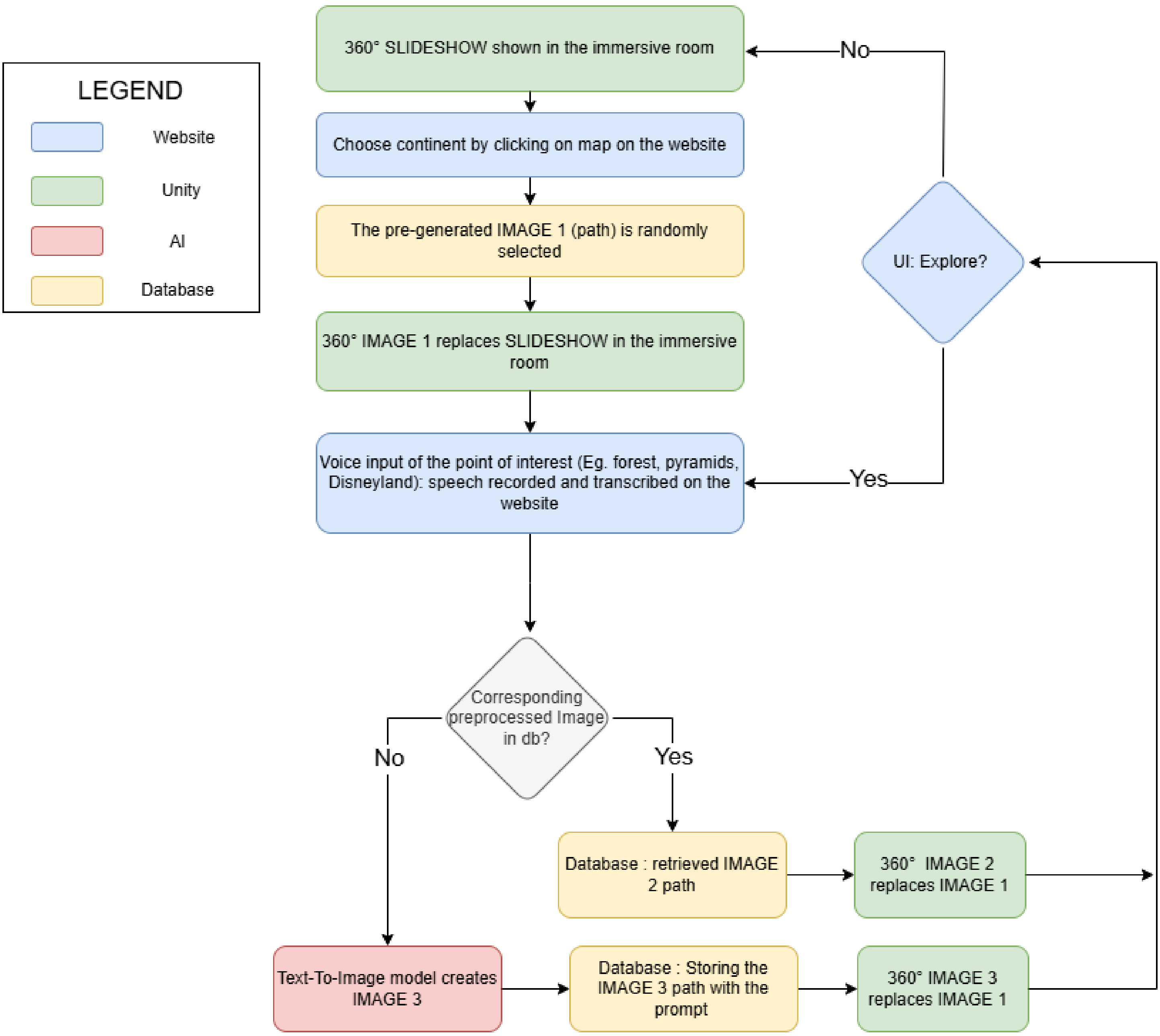
1. **Video/Animation Limitations**:
   - **Problem:** Combining videos or animations with free models within an acceptable waiting time (under 2 minutes) is not feasible.
   - **Solution**: Engage the customer in discussions about their preferences:
     - Free models → static images.
     - Paid models → animations or videos.
   - **Outcome**: Default to static images unless the customer opts for paid solutions.

2. **Waiting time limitations:**
   - **Problem**: Using serverless inference APIs (like Hugging Face) caused unpredictable waiting times (5–40 minutes) due to server availability.
   - **Solutions**:
     - Switch to running smaller, faster models directly on our machines.
     - Pre-generate a library of images featuring famous landmarks.
   - **Outcome:** On-the-spot image generation takes under 2 minutes, and pre-generated images ensure zero wait time in many cases.

3. **AI Accuracy Concerns**:
   - **Problem:** AI-generated images may lack precision and cannot replace traditional educational study trips.
   - **Solution**: Redefine the educational concept through collaboration with the customer:
     - **Creative Engagement**: Use AI-generated images to spark creativity and curiosity in children through interactive scenarios.
     - **Critical Thinking Development**: Foster critical thinking and digital literacy by teaching children to analyze and evaluate AI-generated content, guided by educators.
   - **Outcome**: AI becomes a tool for interactive, critical, and creative learning rather than a direct study trip replacement.

4. **Inaccurate voice recognition**:
   - **Problem**: The website's voice input tool sometimes misinterpreted user speech before sending it to the API.
   - **Solution**: Add "*Confirm*" and "*Deny*" buttons for users to review their recorded speech.
   - **Outcome**: Users can ensure their prompt is correct before generating an image, reducing errors.

User flow diagram



Website

The website component of our project facilitates user interaction with the immersive room through a multi-page interface, hosted on a local network. This web application leverages the **Flask framework** to handle backend logic, incorporating several key endpoints:

- **/process-request**: This endpoint receives the user-selected continent and transcribed voice input from the frontend, returning the request status and a corresponding message. In the absence of voice input, a random environment from the chosen continent is retrieved from the database and displayed in the immersive room.
- **/generate_environment**: This endpoint establishes a connection to the Elastic Search database, searching for a relevant image based on the voice input and continent. If a matching image is found, its path is sent to the Unity application for display in the immersive room. Otherwise, the AI pipeline functions are utilized to generate a new image, which is then sent to Unity for display.

The **frontend** is built using **Vue.js**, providing a visually appealing and intuitive user interface across multiple pages:

- **Starting view**: A simple page featuring an AI-generated background and a start button that initializes the application.
- **World Map view**: An interactive page displaying a world map, where each continent is a clickable button. Upon selection, the chosen continent's value is stored and transmitted to the subsequent page.
- **Loading view**: A transitional page that appears after continent selection, where an API request is made to retrieve or generate the environment.
- **Voice Recording view**: A page enabling users to provide a unique voice prompt, which is transcribed and sent to the API for environment retrieval or generation.

Database

The database component of our project utilizes Elastic Search to store metadata associated with generated images. For each image, the following information is stored:

- **Image path**: The file path of the generated image is stored in Elastic Search, enabling efficient retrieval and reuse of existing images.
- **Enhanced prompt**: The prompt used to generate the image with the AI models is stored, providing context and facilitating future image retrieval based on similar inputs.
- **Title**: A title is assigned to each image, which is used as the filename, allowing for organized storage and retrieval.
- **Continent**: The continent associated with each image is stored, enabling filtering and retrieval of images based on geographical location.
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This metadata is transmitted to Elastic Search in a CSV form at, ensuring efficient data storage and querying.

The continent and voice input are used to search in the metadata and to show a relevant image. In Elastic Search full text search is used, stop words like “the”, “a”, “and” are excluded and level of fuzziness is allowed in the voice input. Each “hit” or found image receives a relevance score. Metadata “title” gets a relevance score which is higher than the enhanced prompt.
To show the image the score must be at least 8 and the image with the highest score is selected.

If no matching image is found a new image is generated. The resulting image is stored, and its metadata, including the folder path, prompt, title, and continent, is added to the database.

By storing the metadata of each created image in the database, our system can efficiently retrieve and display existing images, while also generating new images as needed to accommodate novel user inputs. This hybrid approach balances computational resources with user experience, providing a responsive and engaging interactive environment.

```
id;image_path;title;continent;enhanced_prompt
1;./images/Africa/mountain_waterfall_Zambia.jpg;mountain_waterfall_Zambia;Africa;Africa,a breathtaking panoramic view
2;./images/Africa/Tuareg_camel_in_Sahara.jpg;Tuareg_camel_in_Sahara;Africa;Tuareg and his camel in the sahara desert
3;./images/Asia-Middle-East/Dubai.jpg;Dubai;Asia-Middle-East;Asia-Middle-East,A beautiful cityscape of Dubai with a su
4;./images/Asia-Middle-East/ancient_city_Jordan.jpg;ancient_city_Jordan;Asia-Middle-East;an ancient city in Jordan, wi
5;./images/Asia-Middle-East/Taj_Mahal_.jpg;Taj_Mahal;Asia-Middle-East;The Taj Mahal is a stunning white marble mausole
6;./images/europe/castle_princess_knight.jpg;castle_princess_knight;Europe;a castle in the middle of a forest with a p
7;./images/europe/a_medieval_castle.jpg;a_medieval_castle;Europe;medieval castle with a princess on a horse and a knig
8;./images/europe/iceberg_white_ice_bear_blue_sky.jpg;iceberg_white_ice_bear_blue_sky;Europe;an iceberg with a white i
```

Artificial Intelligence

This project uses AI to capture user speech input and generate media that matches their requests. It creates an AI-powered pipeline that converts user speech into immersive images displayed in the Unity-based environment.

The AI pipeline has five main stages, each playing a key role in turning user voice input into a cohesive and engaging visual experience:

1. **Data Collection and Database Connection**: We start by capturing user voice input through a website interface and transcribing the audio into text using a Vue.js library. The transcribed text is the foundation for the next stages, providing a written version of the user's spoken words.

2. **Text-to-Text Processing (Prompt Enhancement and Profanity Filtering)**: In this stage, we refine and enhance the transcribed text using a specified continent as context. We optimize the prompt to improve performance in the text-to-image generation model and filter out any profanity to ensure a respectful output. We use the Google Gemma-2-2b-it model for this task.

3. **Text-to-Image Generation**: The enhanced prompt is then fed into a fine-tuned version of the Stable Diffusion model called Cyber Realistic, running on local machines. This model generates a 360-degree image based on the refined prompt, considering the continent-specific context from the user's voice input. The resulting image is a direct representation of the user's request, transformed into a visual format.

4. **Inpainting (Seamless Environment Generation)**: To create an immersive experience, we apply an inpainting model to the generated image, smoothing out any seams and mismatched edges. We split the image into two parts, swap them, and then re-generate the edges taking into consideration the content of the whole image using the inpainting model. This creates a cohesive environment where the two image halves seamlessly connect.

5. **Upscaling (Image Resolution Enhancement)**: Finally, we use an upscaling model to enhance the image resolution, increasing it to 8K for crisp and clear display in the immersive room environment. This final stage enables the generated image to be displayed in high detail on the walls of the immersive room, providing an engaging and immersive experience for the user. We use the Real ESRGAN model for this task.

Throughout the pipeline, we use various AI models that work together to transform user voice input into a visually stunning and immersive experience. This demonstrates the potential of AI-powered media generation in creating engaging and interactive environments.

Unity

We use the Unity game engine to display the AI-generated images in the immersive room by running an executable file. The images are sent over the network and then used to update the skybox of the environment. This means that the environment can change in real-time based on the user's input.

To make sure the images are displayed correctly, we use a Unity prefab that was provided by the HITLAB team. This prefab is a template that helps us position and render the images in the right way. By using it, we can ensure that the images are aligned properly and look good in the Immersive Room.

User testing

10/01/2025 Immersive room: image resolution and connectivity

Purposes of the test:
1. Evaluate the resolution of the image.
2. Connect our laptops to Immersive Room PC.

Present :
Thibaut De Tandt, Tom De Cavele, Viola Nguyen, Atvars Apinis, Isabelle Deslé

Results :
1. Resolution must be set to 8k for each Texture2D object in Unity. By default, they are only 2k.
2. Instead of a 5G adapter, the pc will be connected to the internet via an ethernet cable, and the laptop can then send images to it over Wi-Fi.

14/01/2025 Immersive room: inpainting, connectivity

Purpose of the test:
1. Evaluate the image inpainting.
2. Test the connection to the Immersive Room PC.

Present :
Thibaut De Tandt, Viola Nguyen, Atvars Apinis, Isabelle Deslé

Results:
1. To send images to the Immersive Room PC, we need to either turn off the firewall or enable a custom rule to open the necessary ports.
2. We can set the IP address to "all" to receive the images.

20/01/2025 Immersive room: pre-generated images, webservice

Purpose of the test:
Test the generated images, audio, waiting game, slide show, webservice on Viola’s smartphone

- Instead of fun facts, 9 pictures of the MCT robots will be overlaid on the image. This will engage the users while they have to wait for the new picture to generate.
- Test ambient music and sounds in the Immersive Room.
- Webservice is accessible via smartphone, layout is consistent and pretty.

Present :
Thibaut De Tandt, Viola Nguyen, Atvars Apinis, Isabelle Deslé

Results:

- The text box for the game explanation needs to be bigger to be readable.
- Elastic Search queries need to be tweaked.
- Prompt enhancer needs adjustment.
- Make the Text-To-Image model place the focal point of the image on the side, so the inpainting model doesn’t distort it.

22/01/2025 Immersive room: audio, minigame and image generation

Purpose of the test:
Test the audio and the “Find The Robot” game during image generation and ask Thibaut for his voice input.

Present :
Thibaut De Tandt, Viola Nguyen, Atvars Apinis, Isabelle Deslé

Results:

- Robot game works.
- We need to avoid putting the robots over the opening of the Immersive Room.
- Ambient sounds play, and are associated with the image (if the image is not generated on the spot).
- For user generated images, the audio from the last shown pre-generated image keeps playing.
- Research if possible to add automatically an appropriate category for new generated images. (so the images that are generated in real-time have appropriate background music)
- Finding the pre-generated images works better now. Add a continent “other” for all images that are not in a continent or are in more than one continent and adjust query to first search for exact continent and after that for continent “other”.

23/01/2025 Immersive room: sounds, game and image generation

Purpose of the test:
This is the last test before the test with Ilse Meerschaert of Ergo on Friday.
We asked Jordy to give voice input.

Present:
Jordy Verbruggen, Viola Nguyen, Atvars Apinis, Isabelle Deslé

Results:

- "Ambient" catch-all clause for audio works well on real-time generated images.
- AI doesn't generate profanity, violence or inappropriate images because of the negative prompt.
- An A4 with disclaimer and tips for voice input should be available in the Immersive Room.
- All pregenerated Images must be available in the database.
- We will create at least 100 images per continent.

24/01/2025 final test with customer and target audience in the immersive room

Purpose of the test:
Test the total experience, including the use of a pregenerated image, the real-time generation image generation, the game during wait time, the audio. Test users are Japke and Michael, both 13 years old and Ilse Meerschaert and Laura Kyndt of Ergotherapy representing the project customer.

Present:
Japke, Michael, Ilse Meerschaert, Laura Kyndt, Marie Dewitte, Frederik Waeyaert, Thibaut De Tandt, Atvars, Viola, Isabelle

Results:

1. We need a check point table on paper when doing a test or demo in order not to forget necessary setup.
2. Michael and Japke started first hesitantly for the voice input but after a few images they became more outgoing and added more specificity to their voice input:
    - Input "Everest" in the continent Asia showed mountains with snow.
    - Continent North America an input "Spider-Man" showed Spider-Man with the Eiffel tower in the background, because it found our pre-generated image of Spider-Man, which was in France.
    - Input "the highest buildings in the world" showed a newly generated image of a fictive city with skyscrapers.
    - Input "nature" in the continent Canada generated a beaver in a river and Michael expected to see snow.

3. These first inputs by the children were then followed by more specific inputs and therefore better tailored images.
4. Ilse and Laura then simulated the type of conversation they would have with the real target audience.
5. The game during model inference worked well at distracting the children and preventing boredom.
6. The clients accept the minute long inference time.
7. It is unnecessary and potentially dangerous to make changes to the functionality of the project, so we should focus all our efforts on the documentation instead.

Links:

Github

https://github.com/IsabelleDesle/VirtualGlobeTrotter

Trello

ERGO3: Virtual Globetrotter | Trello